

Tasks 1A–1C: Exploratory Data Analysis of Wine Quality

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Overview

This document presents the exploratory data analysis (EDA) conducted as part of Tasks 1A to 1C in the wine quality prediction project. The goal is to investigate how key physicochemical properties — including alcohol and residual sugar — relate to perceived wine quality in red and white wines. This analysis will guide future predictive modelling.

Task 1A: Initial Data Analysis

Quality Distribution – Red Wine

The first analysis focused on the distribution of the `quality` variable in the red wine dataset. The values range from 3 to 8, with most samples concentrated around 5 and 6.

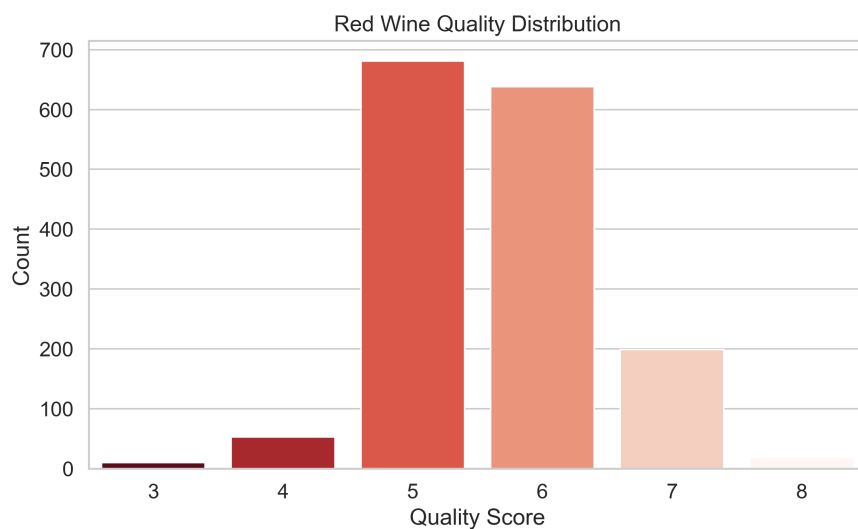


Figure 1: Distribution of Quality Scores in Red Wine Dataset

Observation: The majority of red wine samples are rated 5 or 6 in quality. Higher scores (7 and 8) occur less frequently, and very few samples are rated below 4.

Quality Distribution – White Wine

The same distribution analysis was applied to the white wine dataset.

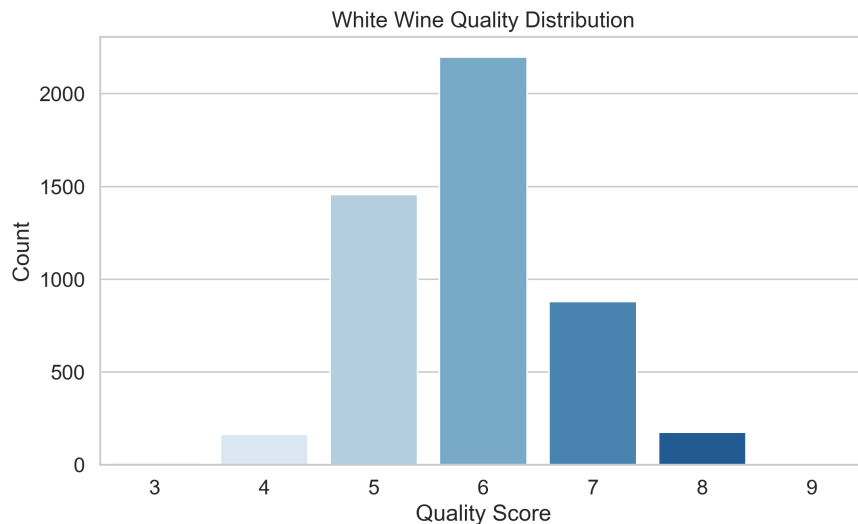


Figure 2: Distribution of Quality Scores in White Wine Dataset

Observation: Like red wine, white wine samples mostly fall in the 5–6 quality range. However, white wine has a wider spread, with a noticeable number of samples rated 7 or 8, and more rated 4 as well.

Red vs White Comparison

To compare the two datasets, the quality distributions were plotted side-by-side.

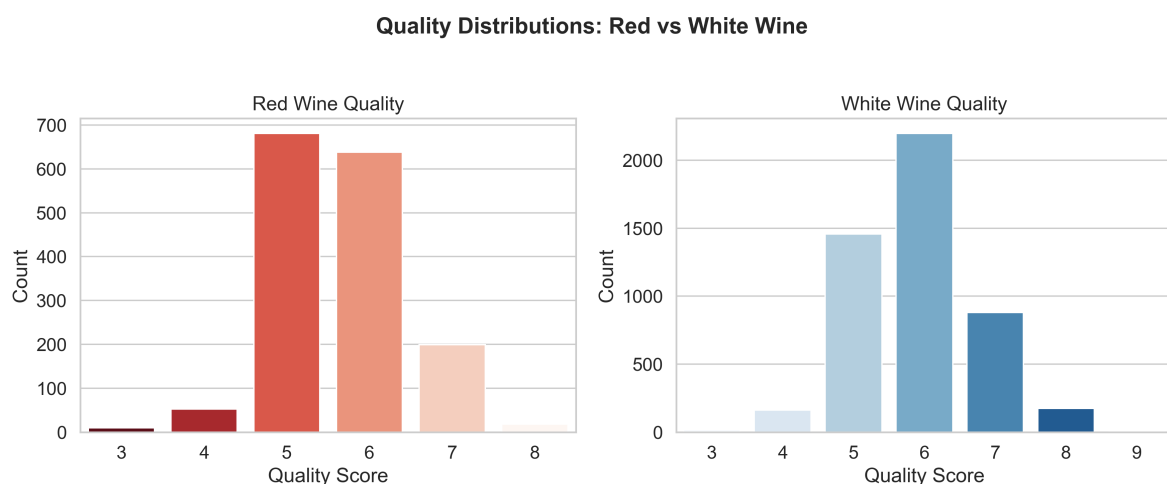


Figure 3: Side-by-Side Distribution of Red and White Wine Quality

Interpretation:

- Red wine quality scores are tightly clustered around 5 and 6.
- White wine exhibits more diversity, with higher and lower quality ratings more common.
- This variation may be influenced by grape types, fermentation techniques, or sample size (white wine has over three times more records).

Task 1B: Alcohol Content and Wine Quality

Overview

This task examines whether alcohol content is associated with perceived quality. Alcohol levels were categorised as **low**, **mid**, or **high** based on the dataset's mean and standard deviation:

- **low**: below (mean – standard deviation)
- **mid**: within one standard deviation of the mean
- **high**: above (mean + standard deviation)

The new variable `alcohol_cat` enables comparison of quality across these groups.

Red Wine

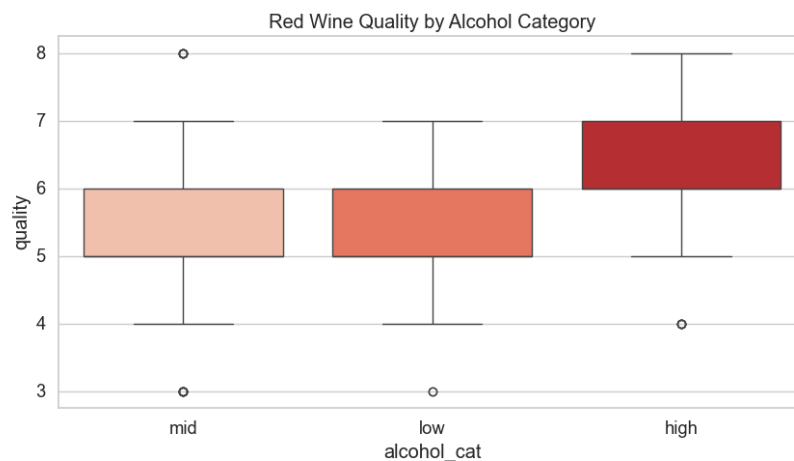


Figure 4: Red Wine Quality by Alcohol Category

Observation:

- Red wines with higher alcohol levels tend to have higher quality scores.
- There is a clear upward shift in the distribution of quality from **low** to **high** alcohol groups.
- Low-alcohol red wines are rarely rated above 6.

White Wine

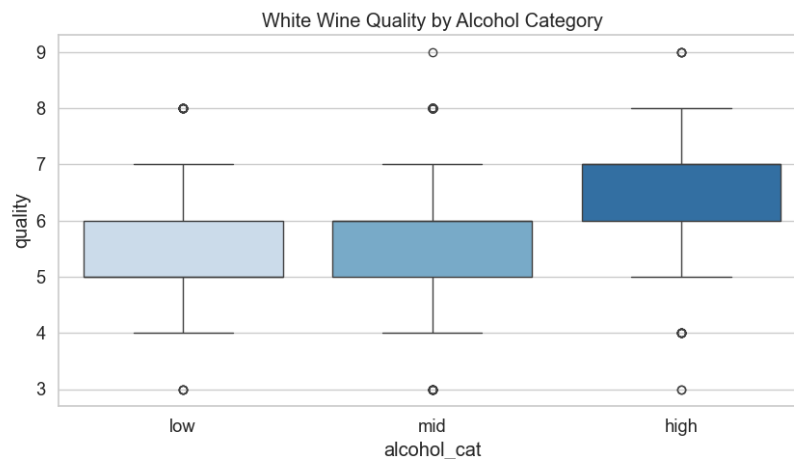


Figure 5: White Wine Quality by Alcohol Category

Observation:

- A similar, though less pronounced, trend is observed in white wine.
- Higher alcohol white wines are rated slightly better, on average.
- The distribution is narrower than in red wines.

Interpretation

- Alcohol content is positively associated with wine quality for both types.
- This may be due to alcohol signalling better fermentation or grape maturity.
- `alcohol` appears to be a strong candidate for predictive modelling.

Task 1C: Residual Sugar and Sweetness Classification

Overview

This task explores whether residual sugar affects perceived quality. A new binary feature `isSweet` was created by splitting each dataset at the median residual sugar level:

- `isSweet` = 1 if residual sugar $\geq$ median
- `isSweet` = 0 otherwise

This provides a roughly balanced classification of wines into “dry” and “sweet”.

Red Wine

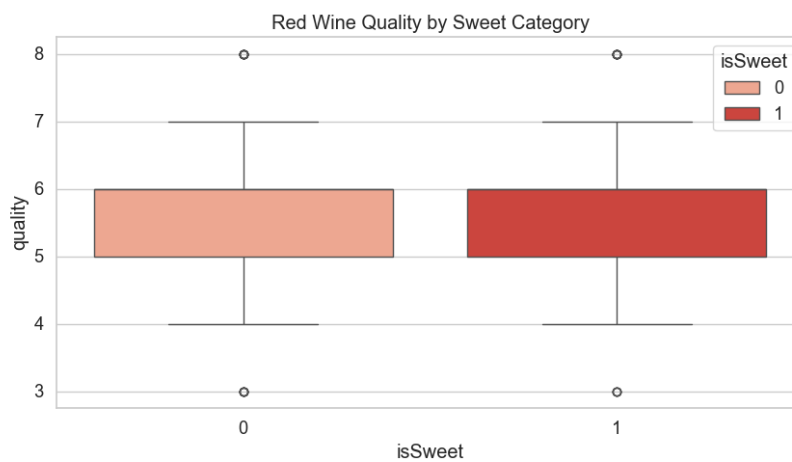


Figure 6: Red Wine Quality by Sweetness (`isSweet`)

Observation:

- Sweet red wines (`isSweet`=1) score slightly lower on average.
- Quality distributions for dry and sweet red wines overlap substantially.

White Wine

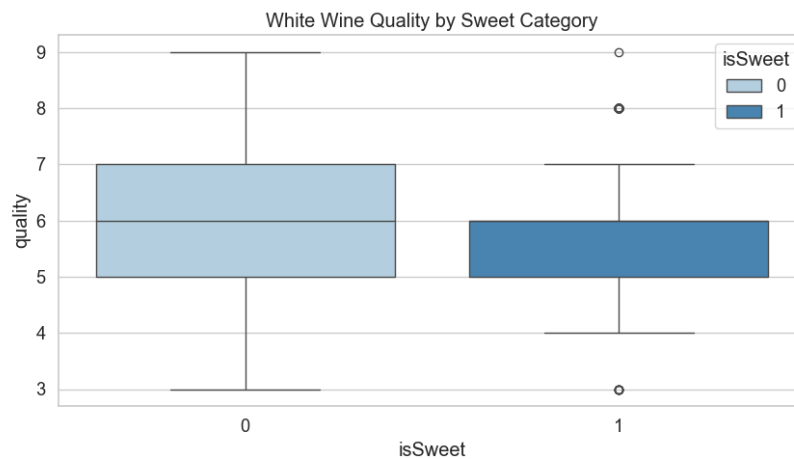


Figure 7: White Wine Quality by Sweetness (`isSweet`)

Observation:

- White wine shows a clearer quality distinction.
- Dry wines tend to have higher scores than sweet ones.
- Sweet white wines are more frequently rated 5 or lower.

Interpretation

- Residual sugar has a weak negative correlation with quality.
- This suggests that drier wines may be perceived as higher quality, possibly due to taste expectations.
- `isSweet` could be a useful feature in predictive models, though its influence is less than alcohol.